

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	29 September 2026
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

The following functional requirements define the core behaviour of the proposed ServiceNow-based metro ticketing solution. They are derived from the actual project workflow: Service Catalog booking, fare logic, Flow Designer automation, structured ticket storage, QR-based digital ticketing, request tracking, and role-based administration.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Metro Ticket Booking	Provide a Service Catalog item for passengers to book a metro ticket. Capture Source, Destination, Passenger Type and Number of Tickets. Validate mandatory journey inputs before submission.
FR-2	Station & Journey Selection	Present valid source and destination station options. Prevent incomplete or invalid route selections. Use station mapping as an input to fare calculation.
FR-3	Automated Fare Calculation	Calculate the fare from configured source-destination / fare-band logic. Apply ticket quantity to calculate the final amount. Display the fare before or during booking confirmation. Maintain consistent fare output for repeated valid inputs.
FR-4	Request Submission & Workflow	Create the ServiceNow request / requested item after submission. Trigger the configured Flow Designer process. Use request variables as inputs to downstream automation. Create required tasks dynamically where applicable.
FR-5	Booking Data Storage	Map catalog variables to fields in the custom Metro Database table. Store structured booking / ticket information for operational use. Maintain linkage between the request context and metro ticket record.
FR-6	QR Digital Ticket Generation	Generate or prepare a QR-based digital ticket after successful processing. Encode the required booking/ticket information. Display the QR ticket through a Service Portal-compatible widget / spModal flow. Ensure the displayed ticket corresponds to the stored booking data.
FR-7	Request & Ticket Status Tracking	Allow the passenger to view the submitted request status. Reflect the progression of the booking workflow. Provide clear confirmation when the ticket is ready / generated.
FR-8	Ticket Verification Support	Allow station staff / authorized operations users to access the relevant ticket information. Support QR-based validation against the booking record. Present sufficient journey data for verification.
FR-9	Role & Access Management	Use roles and ACLs to restrict access to protected booking data. Separate passenger/requester access from operational/admin access. Allow authorized administrators to configure and maintain the solution.
FR-10	UI Behaviour & Dynamic Form Logic	Use UI Policies for conditional visibility / mandatory behaviour. Use Catalog Client Scripts only where dynamic population or essential client-side behaviour is required. Keep the booking experience clear and minimal.
FR-11	Workflow Monitoring & Troubleshooting	Provide Flow Designer Execution Details for tracing failed actions and condition paths. Support debugging of variable population, fare mismatches and automation failures through logs / execution information. Allow administrators to verify mappings and outputs during testing.

FR-12	Notifications & Handoff Documentation	Support configured booking/request notifications where used. Document catalog, flow, table, ACL, testing and deployment setup. Provide sufficient setup guidance to recreate the solution in another Personal Developer Instance.
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Non-functional Requirements:

The non-functional requirements define the quality expectations for usability, security, reliability, performance, availability and scalability. These requirements are expressed at the application design level; infrastructure-level guarantees remain dependent on the ServiceNow platform and the deployed instance.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The passenger-facing catalog form should be simple, readable and guided. Required fields, dynamic visibility and validation should reduce confusion. The QR ticket should be easy to open and present from the portal.
NFR-2	Security	Access to booking and operational data should be governed through ServiceNow roles and ACLs. Users should only access information permitted by their role. Administrative configuration should remain restricted to authorized users.
NFR-3	Reliability	For valid inputs, the same booking data should be transferred consistently from catalog variables through Flow Designer into the Metro Database. Fare results and QR ticket data should correspond to the submitted journey information.
NFR-4	Performance	Normal booking actions should complete without unnecessary manual steps or avoidable client-side processing. Form scripts should be kept minimal, and Flow Designer should be used for maintainable server-side workflow processing.
NFR-5	Availability	The application is expected to be accessible through the deployed ServiceNow instance whenever that instance and required platform services are available. No independent uptime guarantee is asserted by this project.
NFR-6	Scalability	The design should support adding more stations, fare mappings, passenger categories, ticket records and workflow rules without redesigning the full solution. Future payment, email/SMS and analytics functions should be attachable as extensions.
NFR-7	Maintainability	Prefer configuration and low-code components such as Service Catalog, Flow Designer, UI Policies and ACLs. Custom scripting should be limited to necessary behaviour. Setup documentation should support handoff to another administrator/developer.
NFR-8	Data Integrity	Source, destination, passenger details, fare and ticket information must remain consistent across the request and the custom Metro Database record. Validation and mapping errors should be detectable during testing.
NFR-9	Compatibility	The passenger experience should work in the intended Service Portal context. QR display should use a portal-compatible widget/spModal approach rather than depending on backend-only rendering behaviour.
NFR-10	Auditability & Troubleshooting	Administrators should be able to trace workflow execution, inspect condition paths, review stored data, and use available logs/execution details to diagnose failures.

Future / Out-of-Scope Requirements for the Current Version

- Payment gateway integration for UPI, cards or wallets.
- SMS and email delivery of the completed digital ticket.
- WhatsApp-based booking or ticket delivery.
- Travel analytics dashboard and passenger-behaviour analysis.
- Additional mobility integrations such as passes or wider metro ecosystem services.

Requirement Traceability Summary

The critical end-to-end requirement path is: passenger enters journey data -> fare is calculated -> request is submitted -> Flow Designer processes the booking -> data is stored in the Metro Database -> QR ticket is displayed -> authorized station/operations users can verify the ticket. This path also defines the highest-priority testing and troubleshooting scenarios.